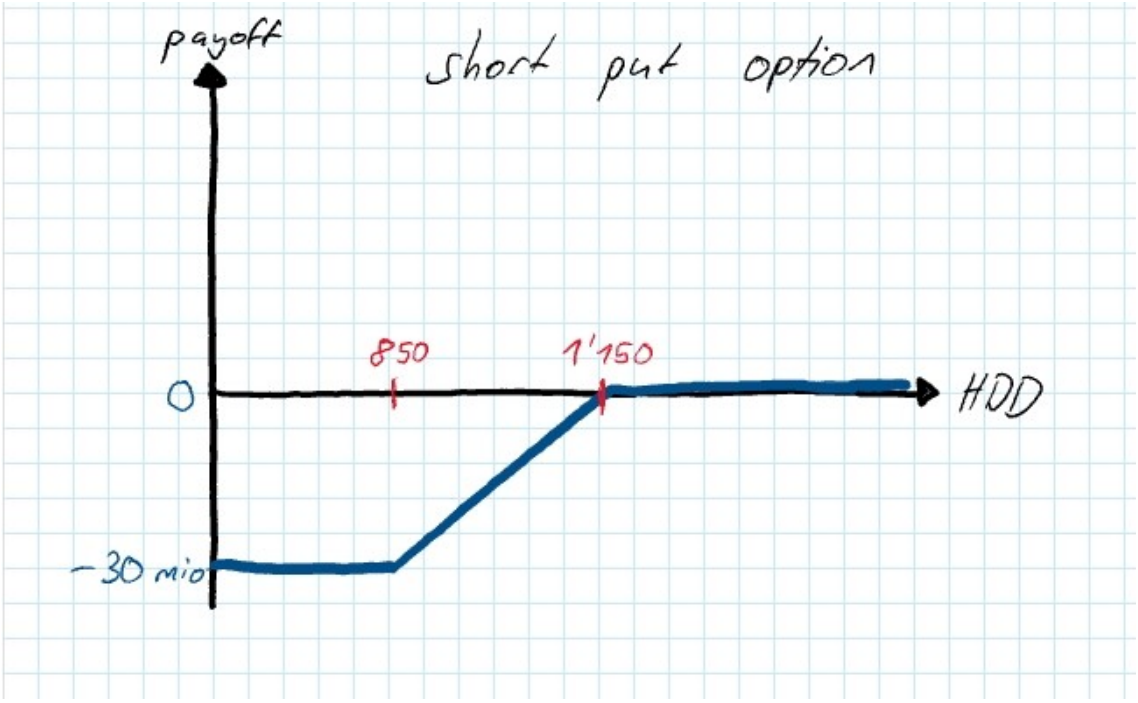
Risk Management Assignment 2

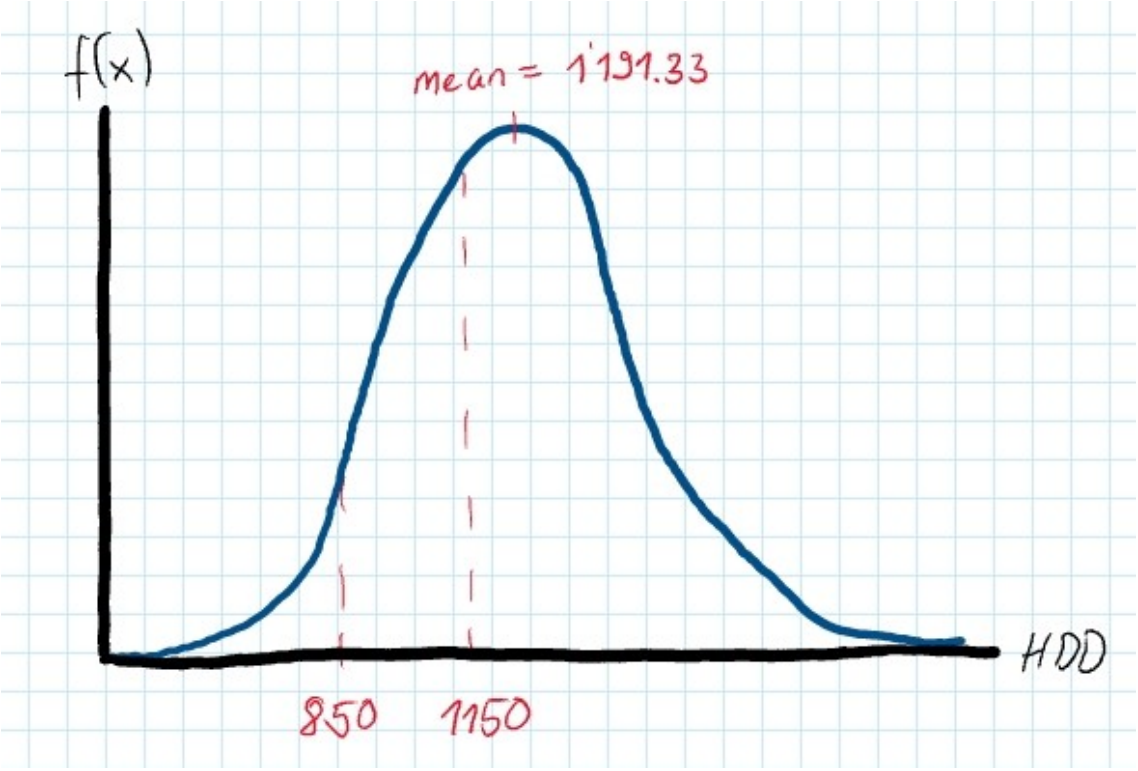
12. 11. 2025

Question 1

Sketch of HDD put (excluding payments to Nephila):



Sketch of index distribution function:



If it's very cold there will be no payment (HDD> 1150). If its too warm then there is a limit to the payment of 30 milions (reached if HDD < 850). In between there will be a payment of 100'000 per HDD. The Index is non-negative.

Question 2

Nephila's pricing rule sets the price as the expected payout plus a return on capital based on the Value-at-Risk (VaR). It's a simple and practical way to make sure expected losses are covered while rewarding investors for taking on risk. The downside is that VaR only looks at losses up to a certain point (like the worst 1%) and ignores how big the losses could be beyond that. A better measure for that would be the Expected Shortfall, which takes those extreme losses into account. But since Nephila's HDD contract has a payout cap of 30 million USD, the potential losses are limited, so VaR and Expected Shortfall give almost the same result. In this case, using VaR still makes sense and keeps the pricing approach straightforward.

To price the option, we assume the HDD index follows a lognormal distribution (since the index is non-negative) and calculate the expected payout and the 99.5% Tail Value-at-Risk (TVaR) to account for extreme outcomes. Following Nephila's pricing rule, we then set the offer price as the expected payout plus a 9% return on capital applied to the TVaR.

For our calculation we followed the pricing rule of Nephila but instead of VaR we used TVaR (or expected shortfall):

$$\text{Offer} = \mathbb{E}[\text{pay}] + r\,(\text{TVaR}_\alpha - \mathbb{E}[\text{pay}]),$$

To calculate TVaR we let the HDD index be lognormally distributed:

$$S \sim \text{Lognormal}(\mu, \sigma^2), \qquad \ln S \sim \mathcal{N}(\mu, \sigma^2).$$

The capped HDD put payoff is

$$\text{pay}(S) = c \cdot \min\{(K - S)_+, U\},$$

with rate c, strike K, and cap range U (cap level $K - U$).

Then we define the VaR_{99}:

$$\text{VaR}_\alpha = c \cdot \min\{\max(K - q_S(1-\alpha), 0), U\},$$

and then calculate TVaR_{99} as follows:

$$\text{TVaR}_\alpha = \frac{1}{1-\alpha}\mathbb{E}[\text{pay}(S) \cdot \mathbf{1}\{\text{pay}(S) \geq \text{VaR}_\alpha\}].$$

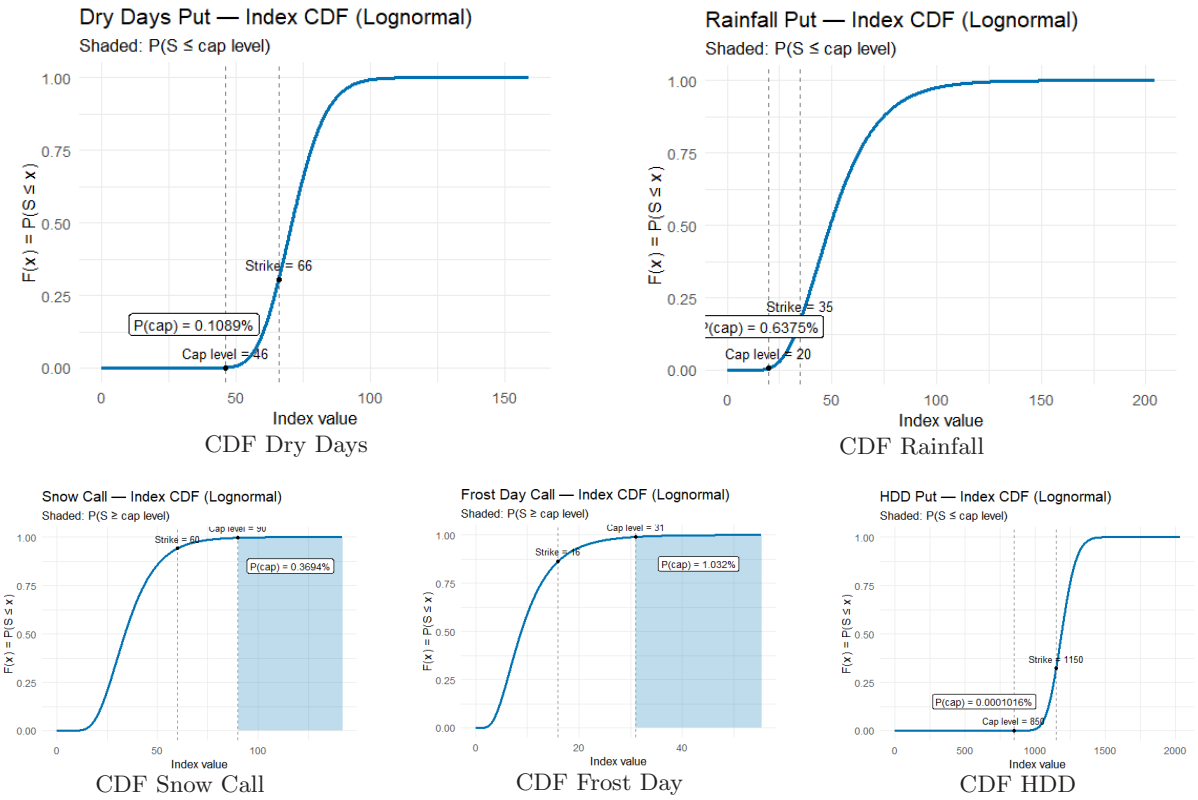
for target return (9%) we get:

$$3'111'135$$

the corresponging R code is in the appendix.

As mentioned above, our calculation confirms that the difference between the price with TVaR and the price with VaR is not very large (3.11 mio. vs. 3.09 mio).

Question 3



CDF Dry Days CDF Rainfall CDF Snow Call CDF Frost Day CDF HDD

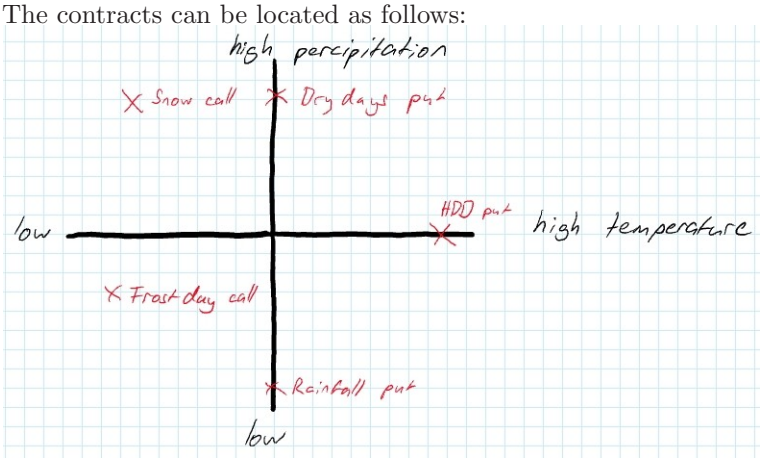
Abbildung 1: Empirical cumulative distribution functions (CDFs) for all weather indicators.

The main reason to impose a cap in weather derivative contracts is to limit the seller's exposure to extreme tail events that, while theoretically possible, would imply catastrophic payouts relative to the premium collected. Weather variables such as temperature and rainfall are bounded but can still show fat-tailed behavior, making extreme but rare outcomes possible. A cap therefore functions as an insurance limit, turning an open-ended payout into a controllable risk profile. It allows Nephila to offer coverage at a feasible price and maintain a consistent return on capital, while buyers still receive protection against meaningful weather deviations.

Assuming the indices follow lognormal distributions with the given means and standard deviations, the calculated probabilities indicate that most caps are conservative. The HDD Put almost never binds (0.0001%), followed by the Dry Days Put (0.11%), the Snow Call (0.37%) and the Rainfall Put (0.64%), which have slightly higher probabilities. Only the Frost Day Call, with a probability of about 1.03%, shows a meaningful chance of reaching its cap.
Overall, the caps effectively protect Nephila against extreme losses, but for most contracts they are positioned so far in the tail that their influence on expected payouts remains limited.

Question 4

The contracts can be located as follows:



Explanation:

- HDD put: Positioned on the right-hand side, it pays out in warm winters with few heating degree days and is largely independent of precipitation.
- Frost day call: Placed on the left, reflecting exposure to cold conditions. When there are no clouds it is typically colder, thus it is placed slightly in the lower precipitation half.
- Snow call: Lies in the upper-left quadrant, dependent on both cold and wet conditions.
- Rainfall put: Pays in dry periods but is mostly independent of temperature.
- Dry days put: Pays if there is high precipitation. Also mostly independent of temperature.

The portfolio of these weather derivatives contains several sources of natural diversification:

- Temperature exposure: Contracts respond to opposing temperature movements. The HDD Put and Dry Days Put are exposed to warm conditions, while the Frost Day Call and Snow Call are exposed to cold weather.
- Precipitation exposure: Some contracts benefit from dry conditions (Rainfall Put, Dry Days Put), others from wet conditions (Snow Call). The Snow Call requires both low temperatures and high precipitation which also provides further diversification as it relies on joint temperature/precipitation dynamics.
- time diversification: The contracts are active in different calendar windows (e.g., HDD and Frost in winter, Dry Days in spring, Rainfall over a longer period), which spreads risk across time.
- Index heterogeneity: The use of different index types like degree-day aggregates, event counts, and precipitation amounts reduces exposure to identical meteorological drivers.
- Correlation structure: From Table 2 of the exercise, we see that correlations vary in sign and magnitude, providing both positive and negative dependencies that help smooth portfolio losses.
- Payout structures and caps: A mix of puts and calls, each with distinct contractual caps, ensures that not all positions exhibit the same tail sensitivity, limiting simultaneous extreme payouts.

Question 5

Since the derivatives in the portfolio exhibit non-linearities due to caps, the Monte-Carlo VaR is more appropriate. Furthermore, the derivatives exhibit joint weather distributions, which also suggests the usage of the Monte-Carlo VaR method. Plus, Delta-VaR assumes normal distributions which seems implausible since none of the indexes can be negative.

Question 6

We estimated the portfolio's risk using a Monte Carlo simulation in R (see code attached). Starting from the historical means, standard deviations, and correlation matrix of the five weather indices, we generated 10'000'000 correlated random outcomes through a Cholesky decomposition of the correlation matrix.

For each scenario, we calculated the payoffs of all contracts based on their triggers, option types, and caps, capturing the nonlinear and limited nature of weather option exposures. We used a log-normal distribution because all of indices are non-negative. Summing these across all instruments produced a simulated loss distribution for the entire portfolio.

From this distribution, we determined the 99% VaR of the portfolio. Our calculated VaR amounts to \$23'543'990, reflecting the portfolio's exposure to extreme adverse weather outcomes.

We can see that the portfolio VaR is significantly lower than the sum of the individual VaRs. This is due to the high diversification potentail as shown in question 4.

Question 7

To allocate the aggregate VaR to individual transactions we use Euler decomposition. We estimated each contract's conditional expected loss given that the total portfolio loss lies near the VaR threshold. This conditional expectation was computed numerically using a Gaussian kernel centered at the VaR point, which smooths losses in a small neighborhood of the quantile to approximate the local gradient. Thus, we have the marginal risk contribution of each option to the portfolio VaR. Using R we get the following allocations

VaR associated with HBB option: 8'449'627
VaR associated with Frost Days option: 3'410'065
VaR associated with Snowfall option: 1'677'827
VaR associated with Rainfall option: 5'733'869
VaR associated with Dry Days option: 4'259'312

For an additional check, we sum up these numbers and we get a number close the portfolio VaR.

Looking at the risk contributions we again see the high diversification potentail of the portfolio as a whole: Each option bears a part of the overall VaR and none of the asset classes has a VaR association of 0. If we look at the share of associated VaR to the portfolio VaR we see that the HDD put option has the largest share (36%) while the Snow Call option has the smalles share (7%). We belief that the small share of the snow call option is due to the fact, that it is exposed to similar risks as Dry Days and Frost Days option thus, providing little additional standalone risk.

Overall, the allocation of the portfolio VaR is consistent with Nephila's internal risk policy, which limits the VaR contribution of any single contract to a maximum of 15 million.

Question 8

To quantify the impact of the portfolio view we look at the calculation of Nephila's reservation price. Now instead of the standalone VaR we assess the contributing VaR to the portfolio VaR. We calculate the difference with the following formula for each option i:

$$\Delta\text{Price}_i = 0.09(\text{VaR}_i^{contribution} - \text{VaR}_i^{standalone})$$

If we subtract the price difference from the standalone reservation price we get the new reservation prices from a portfolio point of view. To evaluate the profitability of each contract, we define profit as the difference between the bid and the reservation price, since the bid represents the amount Nephila receives and the reservation price the minimum amount required to compensate for risk:

$$\text{Profit}_i = \text{Bid}_i - \text{ReservationPrice}_i$$

The following table summarizes the results:

Contract	**Reservation Price old**	**Reservation Price new**	**Reservation Price difference**	**Profitability old**	**Profitability new**
HDD Put	3'091'421.00	2'450'987.4	-640'433.6	238'579.00	879'012.6
Frost Day Call	1'273'759.00	846'589.8	-427'169.2	-73'759.00	353'410.2
Snow Call	483'486.00	266'008.4	-217'477.6	-83'486.00	133'991.6
Rainfall Put	2'951'709.00	2'117'757.2	-833'951.8	-201'709.00	632'242.8
Dry Days Put	2'571'802.00	1'755'960.1	-815'841.9	-71'802.00	744'039.9

So, with the standalone price assessment Nephila gets an expected profitability of its portfolio of -192'177.00 and with the portfolio aggregation the firm will have an expected profitability of 2'742'697.00, so the portfolio aggregation resulted in a clear improvement of expected profitability. This is due to the risk diversification as seen in question 4.

Question 9

Moving from standalone to aggregate portfolio-based pricing allows Nephila to recognise diversification and align deal pricing with its overall 9% capital return target. However, it introduces model dependence, potential cross-subsidisation, and governance complexity. A recomposition of the portfolio would then necessarily lead to a recalculation of the whole portfolio. Portfolio-based pricing is appropriate for large, well-diversified portfolios with stable, validated correlation structures, otherwise a standalone approach remains preferable.

Appendix

Code Ex2

Listing 1: Code Ex2

```r
# --- Inputs ---
mean_x <- 1191.334375; sd_x <- 84.148776
K <- 1150; U <- 300; c_rate <- 100000
alpha <- 0.995

# --- Lognormal parameters ---
sigma2 <- log(1 + (sd_x/mean_x)^2)
sigma  <- sqrt(sigma2)
mu     <- log(mean_x) - 0.5*sigma2
Phi <- pnorm; FS <- function(s) plnorm(s, mu, sigma)
m1  <- exp(mu + 0.5*sigma^2)

# --- Helper functions ---
E_put_uncapped <- function(K_) {
  if (K_ <= 0) return(0)
  a <- (log(K_) - mu)/sigma
  K_*Phi(a) - m1*Phi(a - sigma)
}
E_S_leq <- function(s) m1 * Phi((log(s) - mu - sigma^2)/sigma)

# --- Expected payout (capped) ---
E_pay <- c_rate * (E_put_uncapped(K) - E_put_uncapped(K - U))

# --- VaR_alpha of payout ---
VaR_pay <- function(alpha_) {
  s_q <- qlnorm(1 - alpha_, mu, sigma)
  c_rate * max(0, min(K - s_q, U))
}
VaR_a <- VaR_pay(alpha)

# --- TVaR_alpha of payout ---
TVaR_pay <- function(alpha_) {
  y_a <- VaR_pay(alpha_)
  if (y_a >= c_rate*U) return(c_rate*U)
  s_a <- K - y_a/c_rate
  s_min <- K - U
  tailp <- 1 - alpha_
  if (s_a <= s_min) return(c_rate*U)
  part <- K*(FS(s_a)-FS(s_min)) - (E_S_leq(s_a)-E_S_leq(s_min))
  (c_rate * (U*FS(s_min) + part)) / tailp
}
TVaR_a <- TVaR_pay(alpha)

# --- Offer price with ROC = 9% ---
ROC <- 0.09
Offer <- E_pay + ROC * (TVaR_a - E_pay)

list(Expected_Payout = E_pay, TVaR_0.995 = TVaR_a, Offer_Price = Offer)
```

Code EX 6ff

Listing 2: Code Ex2

```r
## Monte Carlo VaR for Nephila
## ================================
library(Matrix)

set.seed(1)

## 1) Contract & index inputs (means/sds are on the ORIGINAL index scale)
contracts <- data.frame(
  name     = c("HDD Put","Frost Day Call","Snow Call","Rainfall Put","Dry Days Put"),
  type     = c("put","call","call","put","put"),            # trigger direction
  mean_x   = c(1191.334375, 10.25, 36.225231, 52.565122, 71.625),
  sd_x     = c(  84.148776,  5.983849, 13.655766, 19.652364, 10.165565),
  strike   = c(1150, 16, 60, 35, 66),                       # index strike
  caprng   = c(300, 15, 30, 15, 20),                        # range in index units to cap
  maxPay   = c(30e6, 15e6, 15e6, 15e6, 15e6),               # USD max payout
  stringsAsFactors = FALSE
)

## Payout rate implied by cap: rate = maxPay / caprng (USD per index unit)
contracts$rate <- contracts$maxPay / contracts$caprng

## From Nephilas P&L perspective:
##  +1 if Nephila pays (puts),  -1 if Nephila receives (calls)
contracts$sign_nephila <- +1L
```

```r
## 2) Correlation matrix (your table)
Sigma_corr <- as.matrix(read.table(text="
1           0.505897314 0.162730339 0.298301026 0.10104819
0.505897314 1           0.085248955 0.461434846 -0.232804123
0.162730339 0.085248955 1           -0.091077212 0.107691492
0.298301026 0.461434846 -0.091077212 1           -0.518763707
0.10104819 -0.232804123 0.107691492 -0.518763707 1
", header=FALSE))
colnames(Sigma_corr) <- rownames(Sigma_corr) <- contracts$name

## 3) Convert mean/sd -> lognormal parameters (meanlog, sdlog) per index
sigma2 <- log(1 + (contracts$sd_x / contracts$mean_x)^2)
sdlog  <- sqrt(sigma2)
meanlog <- log(contracts$mean_x) - 0.5 * sigma2

## 4) Monte Carlo settings
N <- 10000000   # number of scenarios (increase for stability)

## 5) Build correlated standard normals via Cholesky
##    (chol returns upper-triangular U with Sigma = U'U; use %*% t(U))
U <- chol(Sigma_corr)                          # upper triangular
Z <- matrix(rnorm(N * nrow(contracts)), N)   # iid N(0,1)
Zc <- Z %*% U                                  # correlated N(0, Sigma)

## 6) Transform to lognormal indices: S_i = exp(meanlog_i + sdlog_i * Zc_i)
S <- sweep(Zc, 2, sdlog, `*`)
S <- sweep(S, 2, meanlog, `+`)
S <- exp(S)
colnames(S) <- contracts$name

## 7) Price each capped contract per scenario
##    Put:   payoff = rate * pmin( pmax(strike - S, 0), caprng )
##    Call:  payoff = rate * pmin( pmax(S - strike, 0), caprng )
pay_mat <- matrix(0, nrow=N, ncol=nrow(contracts))
for (j in 1:nrow(contracts)) {
  K  <- contracts$strike[j]
  Uc <- contracts$caprng[j]
  rt <- contracts$rate[j]
  if (contracts$type[j]=="put") {
    intrinsic <- pmax(K - S[,j], 0)
  } else {
    intrinsic <- pmax(S[,j] - K, 0)
  }
  pay_mat[,j] <- rt * pmin(intrinsic, Uc)
}
colnames(pay_mat) <- contracts$name

## 8) Portfolio loss for Nephila: puts are losses (+), calls are gains (   )
loss_mat  <- sweep(pay_mat, 2, contracts$sign_nephila, `*`)  # effectively identical to pay_mat
port_loss <- rowSums(loss_mat)   # total loss (USD)
i_check <- 420
indices_i  <- S[i_check, ]
losses_i   <- loss_mat[i_check, ]
total_loss <- port_loss[i_check]

cat("\n--- Scenario", i_check, "---\n")
print(round(indices_i, 4))     # simulated index realizations
print(round(losses_i, 2))      # loss per contract (USD)
cat("Total portfolio loss (Nephila):", round(total_loss, 2), "USD\n")

## 9) Compute VaR and TVaR (Expected Shortfall)
alpha  <- 0.99
VaR_99  <- as.numeric(quantile(port_loss, probs = alpha,  type = 7))
TVaR_99 <- mean(port_loss[port_loss >= VaR_99])

alpha2 <- 0.995
VaR_995  <- as.numeric(quantile(port_loss, probs = alpha2, type = 7))
TVaR_995 <- mean(port_loss[port_loss >= VaR_995])

ELoss   <- mean(port_loss)           # expected portfolio loss (all legs contribute +)
comp_EL <- colMeans(loss_mat)        # component expected losses (all positive)

## sanity checks
max_loss_theory <- sum(contracts$maxPay)   # = 90e6
list(
  Expected_Portfolio_Loss_USD = ELoss,
  VaR_99_USD  = VaR_99,   TVaR_99_USD  = TVaR_99,
  VaR_995_USD = VaR_995,  TVaR_995_USD = TVaR_995,
  Component_Expected_Loss_USD = setNames(comp_EL, contracts$name),
  Theoretical_Max_Loss_USD    = max_loss_theory,
  Empirical_Max_Loss_USD      = max(port_loss),
  Gap_To_Bound_99             = max_loss_theory - VaR_99,
  Gap_To_Bound_995            = max_loss_theory - VaR_995
)
```

```r
## replace your current sign line with this (place before step 8)
contracts$sign_nephila <- ifelse(contracts$type == "put", +1L, -1L)


# Euler allocation for VaR using a Gaussian kernel around the VaR point
# C_j = E[L_j | L     VaR]  (so that sum_j C_j     VaR)
euler_var_kernel <- function(port_loss, loss_mat, VaR, bw = NULL) {
  stopifnot(is.numeric(port_loss), is.matrix(loss_mat), length(port_loss) == nrow(loss_mat))
  n <- length(port_loss)

  # Bandwidth (Silverman) unless provided
  if (is.null(bw)) {
    sL <- stats::sd(port_loss)
    if (!is.finite(sL) || sL == 0) stop("Degenerate loss distribution; sd(port_loss)=0.")
    bw <- 1.06 * sL * n^(-1/5)
  }

  z  <- (port_loss - VaR) / bw
  kw <- exp(-0.5 * z^2) / sqrt(2*pi)      # Gaussian kernel; 1/bw cancels in ratio

  if (sum(kw) == 0) stop("Kernel weights are all zero. Increase 'bw'.")

  # N a d a r a y a Watson  conditional expectation at VaR
  num <- as.numeric(t(loss_mat) %*% kw)   # length = n_instruments
  den <- sum(kw)

  contrib <- num / den                    # Euler contributions
  structure(list(
    contrib   = contrib,
    share     = contrib / VaR,
    check_sum = sum(contrib),
    bw        = bw
  ), class = "euler_var")
}

# ----- Use it on your results -----

# After (re)building loss_mat, port_loss and computing VaR_99, VaR_995:
res99  <- euler_var_kernel(port_loss, loss_mat, VaR = VaR_99)
res995 <- euler_var_kernel(port_loss, loss_mat, VaR = VaR_995)

# Pretty print
alloc99 <- data.frame(
  Instrument = colnames(loss_mat),
  VaR_99_Contribution_USD = res99$contrib,
  Share = res99$share
)
alloc995 <- data.frame(
  Instrument = colnames(loss_mat),
  VaR_995_Contribution_USD = res995$contrib,
  Share = res995$share
)

print(alloc99)
cat(sprintf("Check (sum contrib     VaR_99): %.2f vs %.2f\n", res99$check_sum, VaR_99))

print(alloc995)
cat(sprintf("Check (sum contrib     VaR_995): %.2f vs %.2f\n", res995$check_sum, VaR_995))


mean_payoff <- c(1690521,539684,115004,1601709,1372622)

VaR_associated <- c(
  8449627,   # HDD option
  3410065,   # Frost Days option
  1677827,   # Snowfall option
  5733869,   # Rainfall option
  4259312    # Dry Days option
)

# Add to contracts data frame
contracts$VaR_associated <- VaR_associated
contracts$mean_payoff <- mean_payoff

# Compute: -(mean_x + (0.09 * VaR_associated))
contracts$adj_value <- contracts$mean_payoff + 0.09 * contracts$VaR_associated

# Display results
contracts[, c("name", "mean_payoff", "VaR_associated", "adj_value")]

#calculate profitability
bid <- c(3330000,1200000,400000,2750000,2500000)
```

```r
contracts$bid <- bid
reservation_old <- c(3091421,1273759,483486,2951709,2571802)
contracts$reservation_old <- reservation_old

contracts$prof_old <- contracts$bid-contracts$reservation_old
contracts$prof_new <- contracts$bid-contracts$adj_value

contracts$price_diff <- contracts$adj_value-contracts$reservation_old


contracts[,c("name","prof_old","prof_new", "price_diff")]
```